



TOSHIKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHIKENT

FIRST NAME:		LAST NAME:	
ID NUMBER:		SECTION:	
SIGNATURE:		ROOM NUMBER:	

MIDTERM EXAMINATION

COURSE NAME:

PROBABILITY AND STATISTICS

COURSE NUMBER:

MSC3010

EXAMINATION DURATION:

1 HOUR

ALLOWED TO BE USED:

Your personal laptop or/and your personal calculator

Do you use a laptop?

yes ☐ no ☐

If yes, follow the instruction:

(1) Enter zoom and share your screen before the exam starts

(2) Your login in zoom must be in the form ID_SURNAME

(3) If you FINISHED EARLIER ask a proctor to indicate time:

(4) In case of DISCONNECTION ask a proctor to indicate time:

(5) After the time was indicated you must close your laptop.

- Final answers must be written by **only blue or black, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

Do NOT fill	Q1	Q2	Q3	Q4	Q5	Total
max points	6	6	6	6	6	30
your score						

Question 1:

A multiple-choice quiz consists of eleven questions, each with four possible answers of which only one is correct. A student passes the quiz if eight or more correct answers are obtained. What is the probability that a student who guesses blindly at all of the questions will pass the quiz?

11-questions = 11 trials

success is to guess correctly $\Rightarrow p = \frac{1}{4}$

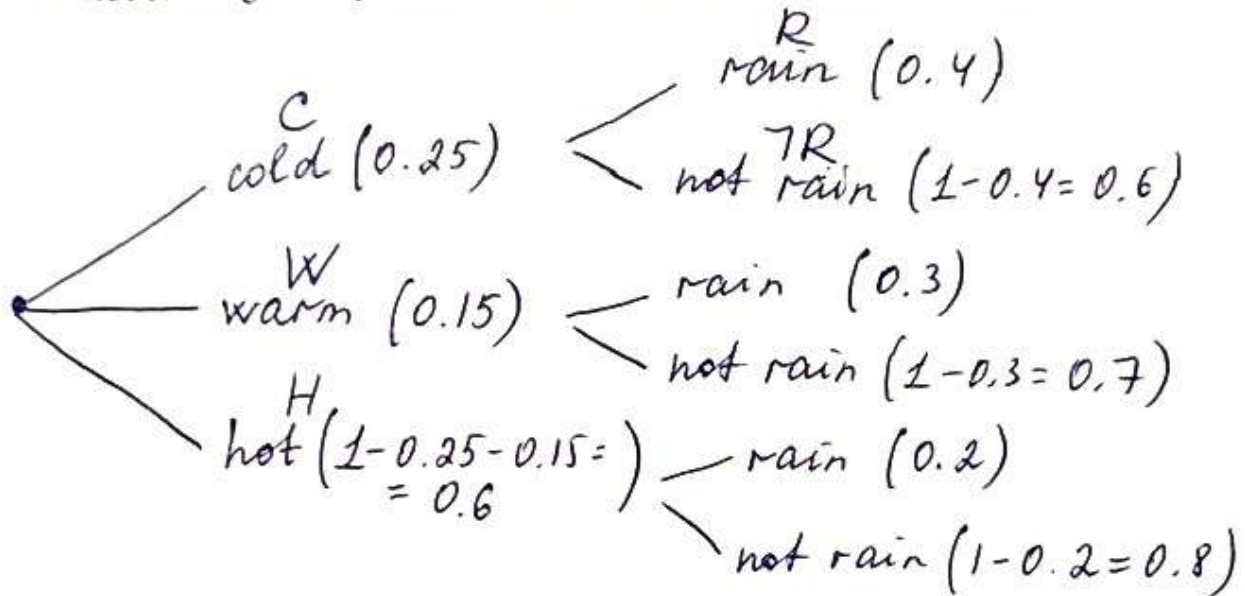
$\Rightarrow$ Binomial $B(11, \frac{1}{4})$

X is number of correct answers

$$\begin{aligned} P(X \geq 8) &= P(X=8) + P(X=9) + P(X=10) + \\ &+ P(X=11) = C_{11}^8 \left(\frac{1}{4}\right)^8 \times \left(\frac{3}{4}\right)^3 + C_{11}^9 \left(\frac{1}{4}\right)^9 \times \left(\frac{3}{4}\right)^2 \\ &+ C_{11}^{10} \left(\frac{1}{4}\right)^{10} \times \left(\frac{3}{4}\right)^1 + C_{11}^{11} \left(\frac{1}{4}\right)^{11} \times \left(\frac{3}{4}\right)^0 = \\ &= \frac{1}{4^{11}} \left(3^3 \times \frac{11!}{8!3!} + 3^2 \times \frac{11!}{9!2!} + 3 \times \frac{11!}{10!1!} + \right. \\ &\left. + \frac{11!}{11!0!} \right) = \frac{1}{4^{11}} \left(\frac{9 \times \overset{5}{\cancel{10}} \times 11 \times 9}{2} + \frac{9 \times \overset{5}{\cancel{10}} \times 11}{2} + 3 \times 11 + 1 \right) \\ &= \frac{4984}{4^{11}} \left(\begin{array}{l} \text{can be left} \\ \text{in this form} \end{array} \right) \approx 0.001188 \end{aligned}$$

Question 2:

The weather on a particular day is classified as either cold, warm, or hot. There is a probability of 0.25 that it is cold and a probability of 0.15 that it is warm. In addition, on each day it may either rain or not rain. On cold days there is a probability of 0.40 that it will rain, on warm days there is a probability of 0.30 that it will rain, and on hot days there is a probability of 0.20 that it will rain. If it is not raining on a particular day, what is the probability that it is cold?



$$P(C | \neg R) = \frac{P(C \cap \neg R)}{P(\neg R)} = \frac{P(\neg R | C) \times P(C)}{P(\neg R | C) \times P(C) +$$

$$+ P(\neg R | W)P(W) + P(\neg R | H)P(H) = \frac{0.6 \times 0.25}{0.6 \times 0.25 + 0.7 \times 0.15 + 0.8 \times 0.6}$$

$$= \frac{0.15}{0.735} = \frac{10}{49} \approx 0.2041$$

Question 3:

A bag contains 200 balls that are either red or blue and either dull or shiny. There are 52 shiny red balls, 89 shiny balls, and 80 red balls. If a ball is chosen at random, what is the probability that it is either a shiny ball or a red ball? What is the probability that it is a dull blue ball?

	red	blue	
shiny	<u>52</u>	37	<u>89</u>
dull	28	<u>83</u>	111
	<u>80</u>	120	<u>200</u>

$$P(S \cup R) =$$

$$= P(S) + P(R) - P(S \cap R)$$

$$= \frac{89}{200} + \frac{80}{200} - \frac{52}{200} = \frac{117}{200}$$

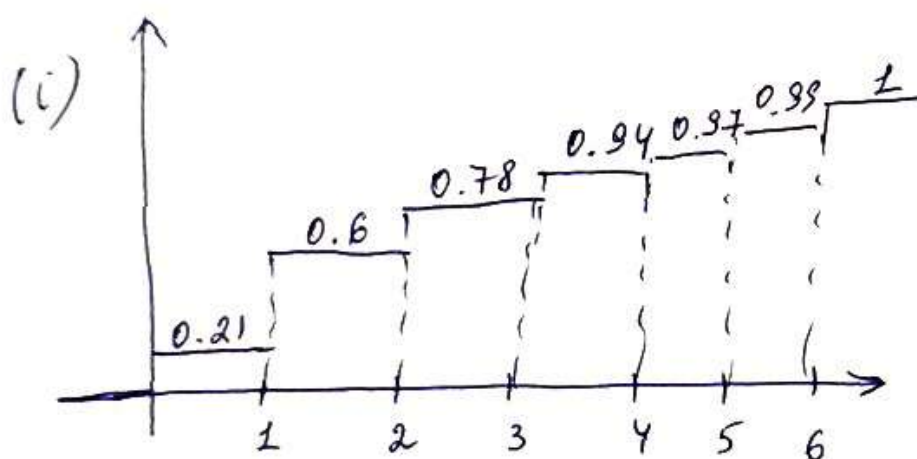
$$P(D \cap B) = \frac{83}{200}$$

Question 4:

The probability mass function of the number of calls taken by a switchboard within 1 minute is given below:

x_i	0	1	2	3	4	5	6
P_i	0.21	0.39	0.18	0.16	0.03	0.02	0.01

- (i) Compute and sketch the cumulative distribution function of the number of calls taken within a minute.
- (ii) What is the expected number of calls taken within a minute?
- (iii) What is the variance of the number of calls taken within a minute?
- (iv) If the numbers of calls taken in two different minutes are independent, what are the expectation and variance of the number of calls taken within one hour?



(iii)

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= 0.39 + 0.72 + 1.44 + \\ &\quad + 0.48 + 0.5 + 0.36 - \\ &\quad - (1.51)^2 = 1.6099 \end{aligned}$$

X_i - number of calls in i th min.

(ii)

$$\begin{aligned} E(X) &= 0 \times 0.21 + 1 \times 0.39 + 2 \times 0.18 + 3 \times 0.16 + \\ &\quad + 4 \times 0.03 + 5 \times 0.02 + 6 \times 0.01 = \\ &= 0.39 + 0.36 + 0.48 + 0.12 + 0.1 + 0.06 \\ &= 1.51 \end{aligned}$$

(iv)

$$E(X_1) + \dots + E(X_{60}) = 60 \times 1.51 = 90.6$$

$$\text{Var}(X_1) + \dots + \text{Var}(X_{60}) = 60 \times 1.6099 = 96.594$$

Question 5:

Calls arriving at a switchboard follow a Poisson process with parameter $\lambda = 4$ per minute.

- (i) What is the expected waiting time between two calls?
- (ii) What is the probability that the waiting time between two calls is less than 10 seconds?
- (iii) What is the distribution (name it) of the time taken for ten calls to arrive at the switchboard?

(i) waiting time is exponential dist.

$$\Rightarrow \frac{1}{\lambda} = \frac{1}{4} \text{ min} = 15 \text{ seconds}$$

$$(ii) P(X < \frac{10}{60}) = P(X < \frac{1}{6} \text{ min}) =$$

$$= \int_0^{1/6} 4 e^{-4x} dx = 4 \left[-\frac{1}{4} e^{-4x} \right]_0^{1/6} =$$

$$= - \left(e^{-2/3} - 1 \right) = 1 - e^{-2/3} \quad \left(\begin{array}{l} \text{can be} \\ \text{left} \end{array} \right)$$

$$\approx 0.4866$$

(iii) ~~the number of events occurring~~
~~within time interval~~

For ten calls this is gamma
distr. with $k=10$ and $\lambda=4$